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9202/1

Paper 1

Mark scheme

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Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from oxfordaqaexams.org.uk

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly Level 2 with a small amount of Level 3 material it would be placed in Level 2 but be awarded a mark near the top of the level because of the Level 3 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available 'any **two** from' is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that 'right + wrong = wrong'.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols / formulae

If a student writes a chemical symbol / formula instead of a required chemical name, full credit can be given if the symbol / formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(...) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

'Ignore' is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do not accept

'Do **not** accept' means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	air		1	AO2 3.4.1 a c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	filtration		1	AO2 3.4.1 c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	any three from: • C contains 2 colours / substances • C contains (colour) A • C does not contain (colour) B • C contains an unknown colour • C is not pure or C is a mixture		3	AO3 3.4.1 d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	(pure food colourings) D and F (reason D and F) have a fixed melting point	allow 80 and 139 allow E and G melt over a range of temperatures	1 1	AO2 × 1 AO3 × 1 3.4.1 a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.5	(conversion) $1.2 \text{ g} = 1200 \text{ mg}$ (mass of food colouring =) $\frac{4.2 \times 1200}{100}$ $= 50.4 \text{ (mg)}$ alternative approach: (mass of food colouring =) $\frac{4.2 \times 1.2}{100} \text{ (1)}$ $= 0.0504 \text{ (g) (1)}$ $(0.0504 \times 1000 =) 50.4 \text{ (mg) (1)}$	allow correct use of an incorrectly / not converted mass value allow correct use of an incorrectly calculated value for mass of food colouring	1 1 1	AO2 3.4.1 a
Total Question 1			10	

Question 2

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	a different number of bonds per carbon atom	allow in diamond each carbon atom is joined to four other atoms allow in graphite each carbon atom is joined to three other atoms	1	AO1 3.2.3 b c
	in diamond all the atoms are joined together or graphite is in layers	allow graphite has delocalised electrons	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	hard		1	AO1 3.2.3 b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	diamond has (many) covalent bonds	allow diamond has a giant covalent structure	1	AO1 3.2.2 f
	(which) are strong bonds		1	
	(which) need large amounts of energy to break		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.4	soft or conducts electricity	allow slippery	1	AO1 3.2.3 c d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	silicon dioxide		1	AO1 3.2.2 f

Total Question 2		8
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Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	wear safety glasses	allow wear gloves	1	AO4 3.3.1.2 b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	carbon dioxide	allow CO ₂	1	AO1 3.3.1.2 b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.3	as time increases the mass (of gas produced) increases		1	AO3 3.8.1 a
	mass increases at a decreasing rate or rate of reaction decreases as time increases	allow the reaction is fastest initially	1	
	the reaction stops at 80 seconds	allow total mass of gas produced = 1.85 g	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.4	(mass of gas produced after 20 seconds =) 1.4 (g)		1	AO2 × 2 AO3 × 1 3.8.1 a
	(mean rate =) $\frac{1.4}{20}$	allow correct use of incorrectly determined mass	1	
	= 0.07 (g/s)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.5	increasing the concentration of the acid increases the rate of the reaction		1	AO1 3.6.4 a 3.8.1 e
	(as) there are more particles in a given volume		1	
	(so) the frequency of collisions increases		1	

Question	Answers	Mark	AO/ Spec. Ref.
03.6	Level 3: The method would lead to the production of a valid outcome. The key steps are identified and logically sequenced.	5–6	AO4 3.3.1.2 b 3.8.1 f
	Level 2: The method would not necessarily lead to a valid outcome. Most steps are identified, but the plan is not fully logically sequenced.	3–4	
	Level 1: The method would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.	1–2	
	No relevant content	0	
	Indicative content: • add (hydrochloric) acid to a conical flask • add magnesium carbonate to the conical flask • measure the mass of magnesium carbonate • use the same mass of magnesium carbonate • measure a fixed volume of (hydrochloric) acid • use the same concentration of (hydrochloric) acid gas collection method • add a stopper (linked to a delivery tube) to the top of the conical flask • start the timer • collect the gas in a gas syringe or - collect the gas in a measuring cylinder • measure the volume of gas given off every 10 seconds or - measure the time taken for a certain volume of gas to be given off mass loss method • place the conical flask on a balance • start the timer • measure the mass of gas given off every 10 seconds or - measure the time taken for a certain mass of gas to be given off • repeat with different sized pieces of magnesium carbonate.		
Total Question 3		17	

Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	53 (%)		1	AO2 3.8.3 a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	OH ⁻		1	AO1 3.5.1 d 3.5.1 f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3	air		1	AO1 3.8.3 a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.4	(temperature) 450 °C	allow a value in the range 400–500 °C ignore high	1	AO1 3.8.3 c
	(pressure) 200 atm / atmospheres	allow a value in the range 150–250 atm / atmospheres ignore high allow 1 mark if both values within range but no units given	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.5	ammonia has a higher boiling point	allow unreacted gases have a lower boiling point	1	AO3 3.8.3 c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.6	(unreacted gases are) recycled	allow reused allow returned to the reactor	1	AO1 3.8.3 c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.7	increases the rate of reaction	allow lowers the activation energy allow ammonia is produced faster	1	AO1 × 1 AO2 × 1 3.8.1 h g
	any one from: • (so) less energy / fuel used • (so) lower temperature is used • (so) a high temperature is used for a shorter time		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.8	(number of molecules = $6.02 \times 10^{23} \times 5$) = 3.01×10^{24}	allow correct use of an incorrect number of molecules	1	AO2 3.6.3 b
	(number of atoms = $3.01 \times 10^{24} \times 4$) = 1.204×10^{25}		1	

Total Question 4		11
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Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	fermentation		1	AO1 3.10.1.2f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	steam	allow water (vapour) allow H ₂ O	1	AO1 3.10.1.2f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.3	contain the same functional group	allow all contain –OH allow have the same general formula	1	AO1 3.10.3.1 a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.4	fuel or solvent		1	AO1 3.10.3.1 b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.5	$ \begin{array}{ccccc} & \text{H} & & \text{H} & \\ & & & & \\ \text{H} & - \text{C} & - & \text{C} & - \text{C} = \text{O} \\ & & & & \\ & \text{H} & & \text{H} & \text{O} - \text{H} \end{array} $		1	AO2 3.10.3.2 a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.6	the more carbon atoms the higher the pH		1	AO3 3.10.3.2 b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.7	do not ionise completely (when dissolved) in water or do not ionise completely in aqueous solution	allow partially dissociate (when dissolved) in water allow partially dissociate in aqueous solution	1	AO1 3.10.3.2 b

Question	Answers				Extra information	Mark	AO/ Spec. Ref.
05.8		C	H	O			AO2 3.6.2 c
	(moles =)	$\frac{1.92}{12}$	$\frac{0.32}{1}$	$\frac{1.28}{16}$		1	
		= 0.16	= 0.32	= 0.08		1	
		$\frac{0.16}{0.08}$ = 2	$\frac{0.32}{0.08}$ = 4	$\frac{0.08}{0.08}$ = 1	allow correct use of incorrectly determined number of moles	1	
	(empirical formula =) C ₂ H ₄ O	allow correct use of incorrectly determined number of moles	1				
	(empirical formula mass =) 44	allow correct use of incorrectly determined empirical formula / mass	1				
$\left(\frac{88}{44} \times \text{C}_2\text{H}_4\text{O} = \right) \text{C}_4\text{H}_8\text{O}_2$		1					

Total Question 5	13
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Question 6

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.1	aluminium is more reactive than carbon	allow aluminium is above carbon in reactivity series	1	AO1 3.3.1.1 d e
	(so) carbon cannot displace aluminium or (so) carbon cannot reduce aluminium oxide		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	(electrolysis) uses large amounts of energy	allow (electrolysis) uses large amounts of electricity	1	AO1 3.3.2 i

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	cryolite		1	AO1 3.3.2 i

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.4	ions are free to move		1	AO1 3.3.2 a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.5	(aluminium ions) gain electrons		1	AO2 3.3.2 d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.6	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$	allow 1 mark for $\text{O}^{2-} \rightarrow \text{O}_2 + \text{e}^-$ ignore attempts at balancing	2	AO2 3.3.2 f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.7	the positive electrode is made of carbon		1	AO1 3.3.2 i
	(which) reacts with oxygen		1	
	to produce carbon dioxide		1	

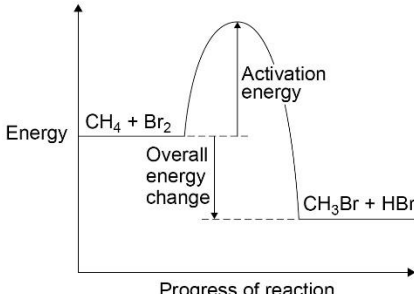
Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.8	$2\text{Al}_2\text{O}_3 \rightarrow 4\text{Al} + 3\text{O}_2$	allow 1 mark for $\text{Al}_2\text{O}_3 \rightarrow \text{Al} + \text{O}_2$ ignore attempts at balancing	2	AO2 3.3.2 i

Total Question 6		13
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Question 7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.1	(a reaction that) transfers energy to the surroundings	allow (a reaction that) releases energy	1	AO1 3.9.1 b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	minimum amount of energy (that) particles must have to react		1	AO1 3.8.1 b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.3	correct shape for exothermic reaction	an answer of: 	1	AO2 3.9.2 d
	labelled products CH ₃ Br + HBr		1	
	labelled activation energy		1	
	labelled (overall) energy change		1	
	scores 4 marks			

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.4	(bond breaking) $((412 \times 4) + 193 =) 1841$	allow correct use of incorrectly calculated values in step 1 and / or step 2	1	AO2 3.9.1 a 3.9.2 f
	(bond making) $((412 \times 3) + 290 + 366 =) 1892$		1	
	$(1841 - 1892 =) (-)51$		1	
	(enthalpy change) $= -51 \text{ (kJ/mol)}$		1	
	Alternative approach (bond breaking) $(412 + 193 =) 605 \text{ (1)}$	allow correct use of incorrectly calculated values in step 1 and / or step 2		
	(bond making) $(290 + 366 =) 656 \text{ (1)}$ $(605 - 656 =) (-)51 \text{ (1)}$ (enthalpy change) $= -51 \text{ (kJ/mol) (1)}$			

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.5	energy must be supplied to break bonds		1	AO1 3.9.2 e g
	(and) energy is released when bonds are formed		1	
	(so) the energy needed to break existing bonds is greater than the energy released forming (new) bonds		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.6	$(M_r \text{ CH}_4 = 12 + (4 \times 1) =) 16$	allow correct use of incorrectly calculated M_r	1	AO2 3.6.1 d 3.6.4 d
	$\left(\text{number of moles of CH}_4 = \frac{1200}{16} \right)$ $= 75$	allow correct use of incorrectly determined number of moles of CH_4	1	
	(number of moles of $\text{H}_2 =$ $75 \times 3 =) 225$	allow correct use of incorrectly determined number of moles of H_2	1	
	(volume of $\text{H}_2 =$) 225×24		1	
	$= 5400 \text{ (dm}^3\text{)}$		1	
Total Question 7			18	